Latent Bridge Matching: Compact Latent Transport for Domain-Level Artistic Transfer

Anonymous submission

Abstract

We present *Latent Bridge Matching* (LBM), a 3.9M-parameter latent-transport model for domain-level artistic transfer. LBM combines training-side endpoint pairing, a style-conditioned latent residual field, and terminal distribution matching in VAE space, enabling minute-scale style-specific customization without adapting a large generative prior. To evaluate genuine style movement in separated art-to-art transfer, we identify an identical-image prior and construct Distinct5-512, a CLIP-separated stress benchmark with an explicit unchanged-image floor (IDT). Under a reviewed same-cost packet with matched roughly two-minute training budgets, LBM reaches transfer CLIP-S 0.6629 at LPIPS 0.3377 and exceeds IDT by +0.0230, whereas same-cost SaMST reaches only +0.0166 at much larger displacement and same-cost SaMAM remains below IDT. On the historical standard benchmark, LBM reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image. Our claim is therefore not universal superiority over all stylization paradigms, but a tighter one that is more relevant to practical deployment: among compact, efficient domain-level style-id models on the reproduced packets studied here, LBM offers a favorable quality-cost frontier while making customized stylization accessible on consumer GPUs.

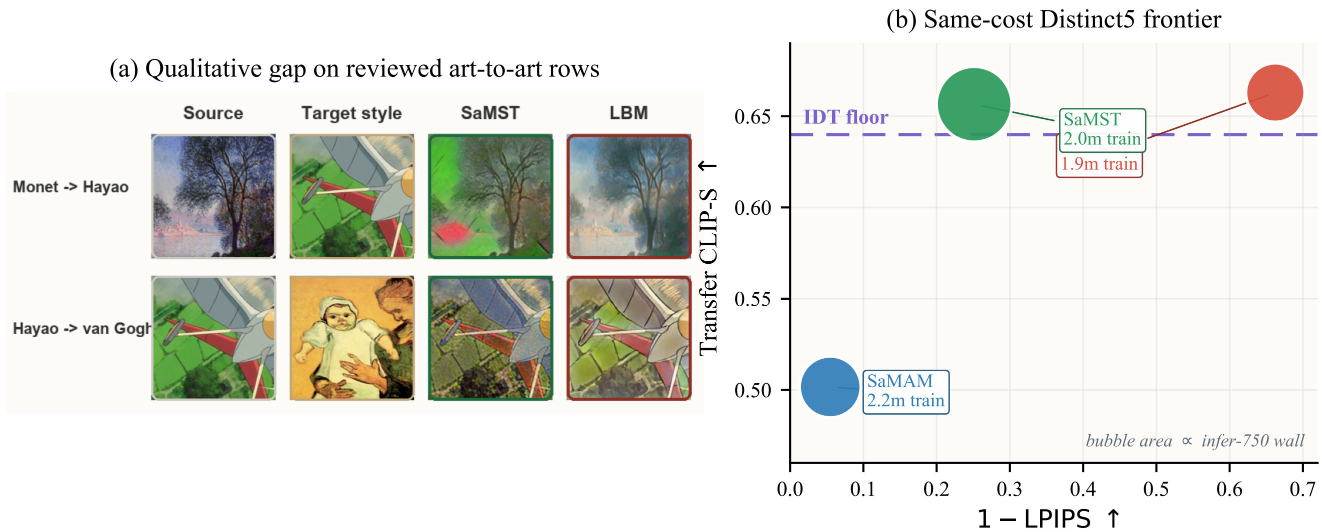


Figure 1: Teaser summary of the paper's two practical claims. In (a), two reviewed art-to-art rows from the historical standard-benchmark packet show the qualitative failure mode of the closest efficient baseline: SaMST can push stronger raw texture, but it does so with muddy grain, color spill, and broken local edges, whereas LBM keeps cleaner target-style movement on the same rows. In (b), the same-cost Distinct5-512 frontier quantifies the matched-budget stress test: higher and further right is better, the dashed line marks the transfer-only IDT floor, and bubble area denotes infer-750 wall time. The LBM point shown here is the explicit matched-budget step-350 packet used for the page-1 comparison, not the strongest retained LBM frontier point near 1.2 minutes. Under matched roughly two-minute training budgets, that packet still occupies the favorable region, same-cost SaMST stays only slightly above IDT at much larger displacement, and same-cost SaMAM remains below IDT.

Introduction

Domain-level artistic transfer asks for a stylized image from a source image and a target style identifier. This is not only an image-generation problem but also a customization problem: the model should not need a reference style image or per-image optimization at inference, and the adaptation path should remain feasible without maintaining a large generative backbone. In practice, three pressures appear together. First, evaluation can fail badly when both source and target domains are artworks: an unchanged source can already obtain a strong target-style score. A method can therefore appear competitive by preserving an art-domain prior, or by making visible but misdirected edits, without moving toward the requested target domain. The first question should be signed target-style movement over the unchanged-image baseline, not raw target-style affinity.

Second, many visually strong stylization pipelines now rely on exemplar-conditioned correspondence or adaptation/steering of large pretrained generators. Those systems can be compelling, but they are awkward if the goal is to build a lightweight style-id model that a small lab, researcher, or creator can retrain and deploy at low cost. For the style-id setting studied here, the practical question is therefore not only *whether* target-style movement happens, but also *who can afford to customize the model and under what adaptation budget*. A useful regime should shift customized stylization from a resource-intensive adaptation routine toward minute-scale learning on consumer GPUs rather than multi-hour tuning on specialized setups.

This failure is not the familiar style/content trade-off. Classical neural style transfer introduced feature reconstruction and Gram statistics (Gatys, Ecker, and Bethge 2016); feed-forward and attention-based arbitrary style transfer improved efficiency and local correspondence (Huang and Belongie 2017; Park and Lee 2019; Deng et al. 2022; Zheng et al. 2024). Recent systems now span reference-guided stylization (Hong et al. 2023; Huang et al. 2024; Zhang et al. 2025; Shang et al. 2025), compact multi-style models selected by style identity (Liu et al. 2024), and diffusion-prior stylizers ranging from trainable prompt-bank adaptation to training-free attention control (Zhang et al. 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024; Jiang et al. 2025; Xu et al. 2025). Across these regimes, automatic metrics can conflate three different events: preserving a source that is already artistic, producing plausible art-domain texture, and moving toward the requested target style. In the style-id setting, the distinction is decisive because inference receives only a domain label.

The representation problem is geometric as well as statistical. VAE latents are compressed feature coordinates with nontrivial local curvature, not RGB pixels. Treating target style as pointwise Euclidean endpoint reconstruction can encourage mean solutions, suppress semantically valid local rearrangements, and blur the line between preserving content and doing nothing. We instead model stylization as controlled latent transport: target-domain examples select where supervision should pull, a style-conditioned vector field decides how to move, and terminal distribution matching checks whether the integrated endpoint has target-style patch statistics.

We propose *Latent Bridge Matching* (LBM), a compact latent-transport model built around that evaluation requirement. In the current mainline, LBM combines (i) training-side endpoint pairing instantiated through a prototype-aware pairing cache and endpoint-mode OMF training; (ii) a style-conditioned latent residual field that executes the requested transport; and (iii) SA-SWD with kinetic regularization, which pulls the Euler-integrated endpoint toward target-style patch statistics while limiting unnecessary displacement. A learnable style spatial prior and semantic cross-attention provide domain-level conditioning without per-image optimization or diffusion sampling. This keeps customization in a lightweight style-id regime: the reported model has 3.91M parameters, reaches a positive Distinct5 operating point within two minutes of training on a consumer GPU, and avoids adapting a large pretrained generator. In this paper, the identical-image floor is used as a split-local reporting check for the proposed Distinct5-512 stress benchmark: style gain is read against that reference rather than from raw target-style affinity alone.

Our contributions are:

- **We formulate domain-level artistic transfer as executable latent transport**, separating style specification from latent execution through training-side endpoint pairing, a time-conditioned latent field, motion regularization, and terminal distribution matching.
- **We identify the identical-image prior in art-to-art evaluation and construct the Distinct5-512 stress benchmark**, where reproduced baselines can visibly edit images yet still remain below the unchanged-image floor; we therefore report Δ_{idt} as a companion measure of genuine target-style movement.
- **We demonstrate a lightweight adaptation regime for customized style models**: in the reviewed setup, LBM reaches a positive Distinct5-512 operating point within two minutes of checkpoint training on a consumer GPU, and the historical standard-benchmark model remains 3.91M with 114 ms/image inference. This lowers the barrier for researchers and creators to build customized style models without adapting a large generative prior.

Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses (Gatys, Ecker, and Bethge 2016). AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics (Huang and Belongie 2017). Attention- and flow-based arbitrary style transfer then pushed local correspondence further: SANet improved style-aware matching, ArtFlow used reversible flows to reduce biased transfer, AdaAttN revisited local attentive normalization, StyTr2 introduced transformer-based content-style interaction, and S2WAT strengthened transformer locality with strip-window attention (Park and Lee 2019; An et al. 2021; Liu et al. 2021; Deng et al. 2022; Xia et al. 2024). More recent reference-guided arbitrary stylization continues to refine the injection path itself: AesPA-Net and AesFA add aesthetic-aware control, Puff-Net emphasizes cleaner content/style feature separation, HSI replaces point-wise attention with a holistic injector, and SCSA focuses on semantic-region consistency in arbitrary semantic style transfer (Hong et al. 2023; Huang et al. 2024; Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025). These works mainly ask how a style exemplar should be fused at inference time. Our problem is different: the main protocol in this paper does not provide a target style image at inference, so the central question is how a domain-level style control should be executed without exemplar-conditioned correspondence.

Efficient multi-style transfer and compact style representations. Multi-style transfer amortizes many styles into one compact model rather than solving a fresh arbitrary-style problem for every target image. SaMST is the closest recent representative: it separates style modeling from style transfer through pluggable style representations and a style-aware transfer network, explicitly targeting the quality-efficiency trade-off for many-style deployment (Liu et al. 2024). This line is the most relevant tokenizer-side baseline for our work. Throughout this paper, "style tokenizer" means a style-id-conditioned control map rather than an image tokenizer, a target-image encoder, or a diffusion-token controller. Our contribution is not another pluggable codebook alone. Instead, we ask whether a compact style representation remains executable after it passes through a latent transport field, so that style-side distinctions survive downstream rendering strongly enough to produce real style movement beyond the unchanged-image baseline.

State-space and Mamba-based style transfer. Recent work also revisits style transfer through state-space backbones. Mamba-ST and SaMam argue that linear-complexity state-space operators can recover global receptive fields more efficiently than quadratic attention, improving the efficiency-quality frontier of arbitrary or many-style transfer (Botti et al. 2025; Liu et al. 2025). These papers are important because they shift the architectural question within compact feed-forward arbitrary and multi-style stylizers from "CNN or transformer" to "which global-mixing primitive is most efficient." Our comparisons to SaMam take this line seriously. The distinction is that LBM does not claim a new global-mixing backbone; its main contribution is training-side and objective-side: endpoint pairing, transport-field execution, and semantic-aligned terminal distribution matching under a compact style-id interface.

Training-free and diffusion methods. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free stylization (Gim, Yoo, and Uh 2024). Diffusion-prior stylization spans both trainable prompt-bank adaptation and training-free control: ArtBank learns an implicit style prompt bank on top of a pretrained diffusion model (Zhang et al. 2024), whereas StyleID and Z* manipulate diffusion attention at inference time without retraining (Chung, Hyun, and Heo 2024; Deng et al. 2024). SMS and StyleSSP further refine this large-prior stylization line through style-matching objectives and improved sampling initialization (Jiang et al. 2025; Xu et al. 2025). These methods can be visually strong, but they generally assume a style reference image and often inherit the inference cost or calibration issues of large generative priors. Our work instead targets compact retrainable latent integration with style-id conditioning rather than iterative diffusion sampling, prompt engineering, or per-style inference-time optimization.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems (Cuturi 2013; Peyré and Cuturi 2019), while the Schrodinger problem connects entropic transport to stochastic path measures (Léonard 2014). We use SWD as a lightweight terminal distribution objective and instantiate the reported Distinct5 headline rows as pairing-cache / terminal-SWD / kinetic OMF with adaptive projection routing through SA-SWD. Earlier exploratory variants in this line also studied OT/Sinkhorn endpoint assignment as a parent objective, and the formal analysis below refers to that machinery only where it is the actual object of study rather than the literal active Distinct5 training loop. Evaluation itself remains unsettled in arbitrary style transfer: prior studies report inconsistent protocols across papers and show that commonly used automatic metrics are style-dependent and require calibration to human preference (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our metric claim is therefore focused rather than universal. For separated art-to-art stress splits of this type, we make the unchanged-image prior explicit through an 'idt' reference and report Δ_{idt} so that target-style movement can be separated from preserved art-domain priors. In the Distinct5 results, the paper-facing ArtFID column is targetwise ArtFID; aggregate ArtFID is retained only as an auxiliary art-domain/identity diagnostic outside the main comparison table. We therefore combine perceptual and distributional measures including LPIPS (Zhang et al. 2018), FID (Heusel et al. 2017), KID (Bińkowski et al. 2018), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), MANIQA (Yang et al. 2022), and ArtFID (Wright and Ommer 2022). This broader metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts or may partially reflect art-domain priors already present before stylization.

Method

Overview

LBM is a latent transport design in VAE space for domain-level artistic transfer. Let $z_0 \in \mathbb{R}^{4\times 32\times 32}$ denote the VAE latent of a content image and let s denote a target style domain identifier. It comprises four coupled components: (i) training-side endpoint pairing that selects candidate target latents without paired supervision; (ii) a time-conditioned transport field $v_\theta(z_t, t, s)$ that realizes the corresponding style-conditioned transport; (iii) kinetic regularization that discourages unnecessary latent displacement; and (iv) SA-SWD terminal matching at the integrated endpoint. At inference, the learned vector field is integrated with a fixed-step solver and decoded by the VAE decoder. In this decomposition, the tokenizer specifies the target region of style space, and the backbone is the executor that realizes the corresponding transport field. Earlier exploratory OT-style selectors are treated only as historical motivation; the active Distinct5 and standard-benchmark rows use endpoint-mode OMF with cache-based endpoint pairing rather than online

Sinkhorn updates.

In the reproduced Distinct5 setting, the active headline rows use endpoint-mode OMF training rather than the historical random-time flow residual. Writing

$$\hat{v} = v_\theta(z_0, 1, s), \qquad \hat{z}_1 = z_0 + \hat{v}, \tag{1}$$

the active Distinct5 objective is

$$\mathcal{L}_{\text{active}} = \lambda_{\text{term}} \mathcal{L}_{\text{term}}(\hat{z}_1) + \lambda_{\text{kin}} \|\hat{v}\|_2^2, \tag{2}$$

where paired target latents for $\mathcal{L}_{\text{term}}$ are selected by the pairing cache and the optional parent flow residual is disabled in the headline rows ($w_{\text{flow}} = 0$). We therefore analyze LBM at the transport-design level and name objective-family differences only when they matter empirically.

Unpaired latent-domain sampling and endpoint pairing

We train directly on precomputed VAE latents without paired supervision. Each minibatch samples content latents $\{z_c\}$ from all source domains and target-style latents $\{z_s\}$ from the requested target domain. Identity pairs are naturally included through uniform target-domain sampling, which stabilizes the learned transport around same-domain mappings.

In the active pipeline, candidate target endpoints are assigned by a prototype-aware pairing cache and selector schedule rather than by online Sinkhorn optimization at every training step. Concretely, for each content latent z_0 and target style s, the cache exposes a selector-restricted candidate set $\mathcal{C}(z_0, s) \subset \{z_s\}$, and the terminal target $\tilde{z}_1$ used by $\mathcal{L}_{\text{term}}$ is drawn from that scheduled set. Earlier exploratory variants also tested an OT-style mini-batch selector over sliced-Wasserstein costs. That historical route is conceptually useful because it motivates coherent endpoint assignment without paired spatial supervision, but it is not the mechanism behind the paper's headline Distinct5 or standard-benchmark results. We therefore treat OT as background motivation rather than as the literal training algorithm used for the main claims.

Time-conditioned latent transport supervision

The core executor is a time-conditioned velocity field $v_\theta(z_t, t, s)$ that realizes style-conditioned latent motion between selected endpoints. In the active training loop, supervision is endpoint-mode OMF rather than a random-time flow-matching residual: the network predicts $\hat{v} = v_\theta(z_0, 1, s)$, forms $\hat{z}_1 = z_0 + \hat{v}$, and is optimized through terminal SA-SWD together with kinetic regularization. At inference, the same field is integrated with a few-step Euler solver. The codebase still contains exploratory flow-style branches, but those branches are disabled in all headline rows ($w_{\text{flow}} = 0$) and are not required to interpret the reproduced results.

Architecture. The time-conditioned backbone uses sinusoidal time embeddings added to the style code. The implementation is a U-Net-like latent trunk with four semantic cross-attention body blocks at 16×16 resolution. The target style is represented by a learnable spatial prior $P_s \in \mathbb{R}^{128\times16\times16}$. Semantic cross-attention injects the style prior into content features, while skip connections preserve spatial structure.

Style tokenizer as executable control

The style-side interface is deliberately separated from the latent renderer. In the main domain-level protocol, the tokenizer receives only a target style identifier and learned style parameters available at inference:

$$c_s = T_\phi(s), \qquad v_\theta = F_\theta(z_t, t, c_s). \tag{3}$$

It does not read a target image or target latent from the current evaluation batch. Target-domain latents remain on the loss side of the graph through training-side endpoint pairing and SA-SWD, but they are not conditioning evidence for T_ϕ. This boundary is important: otherwise training would solve a reference-guided problem while evaluation reports a style-id problem. In this sense, T_ϕ names where transport should go, while F_θ decides how much latent motion is actually needed to get there. Throughout this paper, "style tokenizer" means a style-id-conditioned control map, not an image-token or diffusion tokenizer and not a target-image encoder.

The simplest tokenizer is a per-style code table. Our current implementation generalizes this interface without changing the renderer input contract:

$$c_s = p_s + \gamma \sum_{k=1}^{K} \alpha_{s,k} a_k, \qquad \alpha_s = \text{softmax}(\ell_s/\tau), \tag{4}$$

where p_s is an optional direct prototype, $\{a_k\}$ are shared style atoms, and γ controls the residual atom contribution. In the current mainline, this interface is paired with a prototype-aware latent target queue so that the renderer receives a stable style carrier without changing its single-code input contract. The tokenizer therefore exposes an executable style vocabulary while preserving the compact interface required by the existing renderer.

For execution, the backbone may also use style spatial priors and content-conditioned routing. These modules modulate where a target-style control is applied, not what target style is requested. This distinction matches the empirical diagnosis: the key question is not raw tokenizer capacity in isolation, but whether the style carrier remains executable after it passes through a content-conditioned renderer. Prototype-aware target queues and content-guided routing improve that executed style-content frontier, which is why we phrase tokenizer evidence at the level of *executable representation*: code geometry matters only insofar as generated latent residuals remain distinguishable after the renderer consumes the code.

Complementary roles. The components address distinct subproblems. Training-side endpoint pairing provides candidate target endpoints for unpaired supervision, but not transport dynamics; the broader OT diagnostic explains one historical route to such assignments. Transport supervision learns how to move toward those endpoints, but by itself does not guarantee that the integrated endpoint matches target-domain texture statistics. SA-SWD regularizes this

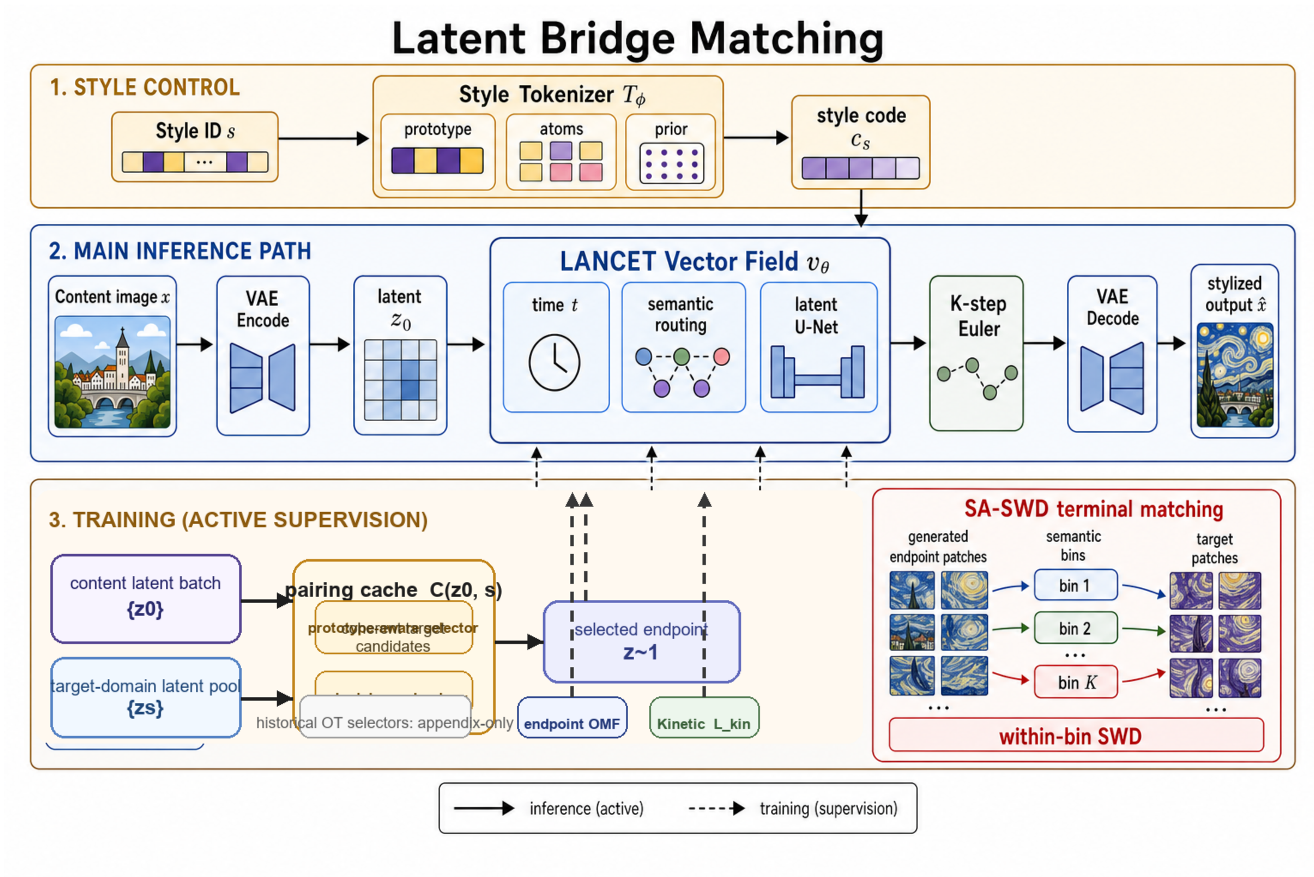


Figure 2: Overview of Latent Bridge Matching. The top band shows style-side control, where the style tokenizer maps the target style identifier into a compact style code. The middle band is the active inference path: content is encoded into a latent, passed through the LBM time-conditioned residual field with semantic routing, integrated by K-step Euler updates, and decoded into the stylized output. The bottom band summarizes the active training constraints used by the headline rows: content latents and target-domain latent pools meet through a prototype-aware pairing cache and selector schedule, which provide the endpoint targets used by endpoint-mode OMF supervision. The active losses are terminal SA-SWD and endpoint kinetic regularization; historical OT/Sinkhorn selector variants are retained only as appendix-side motivation rather than as the main training loop.

endpoint mismatch, while semantic cross-attention provides local routing for where style is written and supplies a routing-derived heuristic for projection selection. The destructive ablations directly verify the roles of terminal distribution matching and kinetic regularization. On the current Distinct5 packet, the dominant gain comes from distributional terminal matching itself, which avoids pointwise latent collapse, while semantic routing remains a tested axis-selection heuristic rather than a separately proven primary driver.

Adaptive-projection terminal distribution matching

In addition to transport supervision, a terminal loss regularizes the Euler-integrated endpoint toward the target style distribution. Let $\Phi_\theta(z_0, s)$ denote the endpoint obtained by integrating v_θ from z_0 with style s. The terminal loss is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1\big(\Phi_\theta(z_0, s),\ \tilde{z}_1\big), \tag{5}$$

where the SWD compares generated endpoint patches with coupled target patches. We use semantic cues to choose SA-SWD projection directions. Let K_s denote the keys produced by semantic cross-attention for target style s. We rank spatial target positions by the average magnitude of K_s and use the corresponding target latent patch vectors as normalized projection directions. This defines the SA-SWD instantiation used in the current mainline: terminal distribution matching is still computed by sorted one-dimensional projections, but the projection axes are adaptively chosen from the same routing signal that injects style into the backbone.

Across the broader LBM family, the training objective can be written as

$$\mathcal{L} = \mathcal{L}_{\text{transport}} + \lambda_{\text{term}}\mathcal{L}_{\text{term}} + \lambda_{\text{kin}}\mathcal{L}_{\text{kin}}, \tag{6}$$

where $\mathcal{L}_{\text{transport}}$ denotes the optional parent transport residual retained only for family-level completeness. In the current Distinct5 headline rows this term is disabled ($\mathcal{L}_{\text{transport}} = 0$, $w_{\text{flow}} = 0$), so the active loss reduces to Eq. (2). The kinetic loss $\mathcal{L}_{\mathrm{kin}} = \mathbb{E}\|v_\theta\|_2^2$ penalizes excessive latent displacement. The primary configuration uses $\lambda_{\mathrm{term}} = 20.0$ and $\lambda_{\mathrm{kin}} = 1.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This decomposition separates family-level template terms from the losses that are actually active in the Distinct5 headline regime.

Metric choice in latent space. The training objective does not treat all latent discrepancies as Euclidean reconstruction errors. The kinetic term uses an L_2 penalty on velocity because it represents path energy, while terminal endpoint matching uses Wasserstein-1 style discrepancies over sorted projections. This separation is deliberate in compressed VAE latents: pointwise endpoint MSE would reward mean solutions and over-penalize semantically valid local rearrangements, whereas SA-SWD preserves distributional supervision without requiring paired spatial correspondence. The current reproduced Distinct5 checkpoints therefore place style pressure on training-side endpoint pairing plus W_1-style terminal matching rather than on latent endpoint MSE. In the measured Distinct5 regime, this design is consistent with repaired endpoint-only controls underperforming the current mainline, without claiming a universal theorem against pointwise endpoint losses.

Theoretical grounding of latent transport. Section provides bounded design-grounding results. Theorem 1 connects the training kinetic loss to the inference-time path action, and Theorem 2 bounds Euler discretization error at $O(1/K)$ for the active few-step field/solver regime used in the current mainline. A separate supplementary note records a historical OT-style selector result only for family-level background.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^{M}$, the empirical terminal objective is

$$\mathcal{L}_{\mathrm{swd}} = \sum_{r\in\mathcal{R}} \frac{1}{M} \sum_{m=1}^{M} \Big\| \mathrm{sort}\big(\mathcal{P}_r(\hat{z}_1)\theta_m\big) - \mathrm{sort}\big(\mathcal{P}_r(z_s)\theta_m\big) \Big\|_1. \tag{7}$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics along semantic routing directions, not whether it matches any particular artwork spatially.

Formal analysis

We provide bounded design-grounding results rather than a single unified theorem for every training variant. Theorem 1 and Theorem 2 analyze the active few-step field/kinetic regime used in the current mainline. A supplementary historical note records an earlier OT-style endpoint-selection result for completeness, but it is not used to justify the current headline claims. Full proofs appear in the supplement. For brevity we state the results and their empirical support.

Notation. Recall the OMF training regime: $\hat{v} = v_\theta(z_0, 1, s)$, $\hat{z}_1 = z_0 + \hat{v}$. At inference, integration follows $dz/dt = v_\theta(z, t, s)$ with K-step Euler: $z_{k+1} = z_k + \Delta t\, v_\theta(z_k, t_k, s)$ where $\Delta t = 1/K$, $t_k = (k + 0.5)\Delta t$. Define the path action $\mathcal{A}(v) = \int_0^1 \mathbb{E}\|v_\theta(z(t), t, s)\|^2\, dt$ and the training kinetic loss $\mathcal{L}_{\mathrm{kin}} = \mathbb{E}\|v_\theta(z_0, 1, s)\|^2$.

Design interpretation. Under bounded temporal drift and Lipschitz continuity, endpoint kinetic regularization acts as a tractable proxy for inference-time path action. Theorem 1 formalizes why a lightweight endpoint penalty can act as a practical proxy for trajectory straightness in the operating regime studied here, without requiring expensive multi-step supervision.

Theorem 1 (Kinetic regularization as a path-energy proxy). Assume that for a fixed content z_0 the velocity norm varies at most δ with t, i.e., $\sup_t \big| \|v_\theta(z_0, t, s)\| - \|v_\theta(z_0, 1, s)\| \big| \le \delta$, and that v_θ is L-Lipschitz in z. Let M bound $\|v_\theta\|$ along the trajectory. Then the path action decomposes as

$$\mathcal{A}(v) = \mathcal{L}_{\mathrm{kin}} + \varepsilon_t + \varepsilon_z, \tag{8}$$

where $|\varepsilon_t| \le 2\delta\sqrt{\mathcal{L}_{\mathrm{kin}}} + \delta^2$ and $|\varepsilon_z| \le LM$. If δ and LM are small, the training kinetic loss provides a tight proxy for the inference-time path energy.

Empirical support. On a historical D0 probe consistent with the current few-step regime (128-step integration): $\mathcal{L}_{\mathrm{kin}} \approx 1017$, $\mathcal{A}(v) \approx 698$, yielding $|\varepsilon_t + \varepsilon_z| \approx 319$. Removing $\mathcal{L}_{\mathrm{kin}}$ (D2) reduces the path length ratio from 0.98 to 0.94, confirming the kinetic term controls path regularity.

Theorem 2 (Euler discretization bound). Under the L-Lipschitz condition on v_θ, the K-step Euler integration error satisfies

$$\|z(1) - z^{(K)}\| \le \frac{C}{K}, \qquad C = \frac{1}{2}(ML_t + M^2 L)\,\frac{e^L - 1}{L}, \tag{9}$$

where L_t bounds the t-derivative of v_θ.

Empirical support. Measured over 160 content latents (D0 model, 256-step reference): $\|z^{(K)} - z^{(256)}\| = \{10.7, 3.80, 1.74, 0.84, 0.40, 0.19, 0.084, 0.027\}$ for $K = 1, 2, 4, 8, 16, 32, 64, 128$; the error halves when K doubles, confirming the $O(1/K)$ scaling. Because VAE latents exhibit non-trivial local curvature, few-step solvers can otherwise accumulate visible drift. Here the measured $O(1/K)$ decay is consistent with the expected solver trend on this historical D0 probe. At $K = 12$ in that probe, the error is < 0.6 relative to the 256-step reference.

A historical OT-style selector result is provided only in the supplement for readers interested in the broader bridge family. We do not rely on that result in the current paper because the active headline rows use cache-based endpoint pairing rather than online OT assignment.

Experiments

Protocol hierarchy and metrics

We organize the evidence into two non-interchangeable result families, with Distinct5-512 as the proposed stress benchmark and the historical strict-750 packet as a standard-benchmark reference surface. This separation is necessary because the same scalar pair, CLIP-style and LPIPS, can be misleading when datasets, image resolution, or baseline training protocols change.

Standard benchmark (historical strict-750 packet). The main legacy protocol uses five domains—photo, Hayao, Monet, Van Gogh, and Cezanne—and evaluates all 5×5 source-target directions with 30 source images per source domain, for 750 generated outputs. This is the protocol used for comparisons to SaMST, S2WAT, StyleID, AdaIN, StyTr2, CAST, AesFA, and AesPA-Net.

Proposed stress benchmark (Distinct5-512). The new stress split uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These five classes were chosen as the most strongly separated subset under a CLIP-style screening pass over WikiArt categories, rather than being hand-picked for LBM. Evaluation again uses all 5×5 directions over 750 generated images. This split complements the historical table with a stronger domain-separation test and an explicit identical-image baseline.

We report CLIP-style (CLIP-S) and CLIP-content (CLIP-C) using CLIP ViT-B/32 embeddings (Radford et al. 2021), together with LPIPS-content and

$$\mathrm{EC} = \text{CLIP-style} \cdot (1 - \mathrm{LPIPS}). \tag{10}$$

For Distinct5-512 we additionally report style gain over the unchanged-image reference,

$$\Delta_{\mathrm{idt}} = \text{CLIP-S}(\text{method}) - \text{CLIP-S}(\text{idt}), \tag{11}$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. We use this quantity to define the metric-stress regime exposed by Distinct5-512: raw CLIP-style should not be treated as sufficient evidence of successful target-style transfer if a method either fails to exceed the identical-image prior ($\Delta_{\mathrm{idt}} \leq 0$) or exceeds it only by moving into a clearly damaged LPIPS region while also deteriorating broad artifact diagnostics. In this paper, CLIP-S is interpreted as target-style affinity, LPIPS as content displacement, and targetwise ArtFID as a target-domain plausibility and artifact diagnostic rather than a direct style-gain score. Aggregate ArtFID is kept only as an auxiliary art-domain/identity diagnostic outside this main Distinct5 comparison. In particular, lower targetwise ArtFID alone does not certify useful stylization on separated art-to-art transfer: its content term rewards source preservation, and its distributional term can reward remaining in a plausible target-domain art manifold even when the requested target-style movement is still below the unchanged-image baseline. EC is used only as a Pareto summary; claims are checked against CLIP-S and LPIPS individually. For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, shallow Gram micro statistics, and KID where available. All standard-benchmark baselines use the same source images, target domains, preprocessing, and identity directions. For arbitrary-style methods, one target-domain reference image is supplied per target style; for style-id LBM, the model receives only the target domain identifier. The 'idt' control is introduced here as a split-specific diagnostic baseline for art-to-art transfer: our claim is that Distinct5 makes explicit the role of the unchanged-image prior on this split, not that any single existing metric should be discarded or universally replaced.

Proposed Stress Benchmark: Distinct5-512

Distinct5-512 is our proposed stress benchmark for separated art-to-art transfer. All five domains come from standard WikiArt classes, and they are selected only because they are among the most separated under a CLIP-style screening pass. In this regime, unchanged output is no longer a plausible success proxy: the *idt* baseline already reaches CLIP-S 0.6801 while doing no transfer at all. Signed style gain is therefore the primary sanity check. Under the shared full 5×5 / 750-output protocol, LBM occupies a low-damage positive-Δ_{idt} region among reproduced compact domain-level operating points. The low-displacement **LBM-Base** point reaches CLIP-S 0.6969 and LPIPS 0.3186 after 1.2 minutes of checkpoint training, while the stronger-style **LBM-Style** point raises CLIP-S to 0.7010 at LPIPS 0.3623 and remains well below the high-damage regime. SaMST-512 pushes raw style higher to 0.7247, but only by entering that regime directly, with LPIPS 0.6255 and targetwise ArtFID 395.7; the retained e5/e15 packet is already near a plateau on transfer CLIP-S and LPIPS, with less than 0.004/0.002 difference between the checkpoints. At the trusted SaMAM-2250 checkpoint, transfer CLIP-S remains below the idt floor even though targetwise ArtFID improves to 146.1 on the full scope and 148.2 on the transfer-only scope. This is the evaluation failure exposed by Distinct5: lower targetwise ArtFID can still coincide with zero or negative target-style gain when a method preserves source structure and stays inside a plausible target-art manifold without moving toward the requested target style. The interpretation is not based on Δ_{idt} alone: Table 2 shows the same failure on representative transfer directions, where SaMAM often edits without consistently moving above the no-op reference and SaMST reaches stronger style only by paying a much larger LPIPS cost. Distinct5 therefore turns the paper's evaluation claim into an explicit operating-point test within this split: target-style movement should be demonstrated relative to idt, not inferred from absolute CLIP-S or ArtFID alone.

Table 3 isolates the recorded operating-point cost under the same Distinct5 packet. It keeps only manuscript rows with closed training records and avoids inventing a shared inference column for rows whose same-grade pure-generation timing is still incomplete. Table 2 then makes the frontier interpretation concrete on representative transfer-only pairs from the reviewed packet: SaMST can produce locally strong target-style jumps, but those gains are not stable across directions and often require much larger displacement than the balanced LBM-Base point, while SaMAM still often edits without moving toward the requested target style.

A paired bootstrap over the 600 IDT-aligned transfer rows gives 95% intervals of $[0.0210, 0.0280]$ for LBM-Base and $[0.0273, 0.0352]$ for LBM-Style, confirming that their positive transfer-only Δ_{idt} values are stable within this reviewed

Table 1: Distinct5-512 proposed stress benchmark. All rows use the same 5×5 evaluation over 750 generated images. 'idt' is the unchanged-image reference copied into every target slot. Δ_{idt} is the full-scope style gain over idt and $\Delta_{\mathrm{idt,tr}}$ is the transfer-only off-diagonal gain. 'tw-ArtFID' denotes targetwise ArtFID. LBM-Base and LBM-Style are the retained low-displacement and stronger-style LBM operating points on this split. SaMAM is reported only at the trusted 2250-step checkpoint; later checkpoints are excluded from the active manuscript path. Timing context is separated into Table 3 and Figure 4 so the main metric table does not mix quality scores with partially closed train/infer records.

Method	CLIP-S↑	LPIPS↓	EC↑	tw-ArtFID↓	Δ_{idt} ↑	$\Delta_{\mathrm{idt,tr}}$ ↑
idt (no-op)	0.6801	0.0000	0.6801	216.5	0.0000	0.0000
SaMAM-2250	0.5811	0.3538	0.3755	146.1	-0.0990	-0.0877
SaMST-512 e15	**0.7247**	0.6255	0.2714	395.7	**+0.0446**	**+0.0558**
LBM-Base	0.6969	**0.3186**	**0.4748**	**122.6**	+0.0168	+0.0244
LBM-Style	0.7010	0.3623	0.4470	157.2	+0.0209	+0.0312

Table 2: Representative Distinct5-512 transfer-only pairs from the same reviewed packet. Cells report CLIP-S / LPIPS for the unchanged-image reference, SaMAM-2250, SaMST e15, and the balanced LBM-Base point. The table is meant as local sanity checking rather than a new aggregate leaderboard. Abbreviations: ER = Early Renaissance, Imp = Impressionism, Min = Minimalism, Roc = Rococo, Ukiyo = Ukiyo-e.

Pair	No-op	SaMAM-2250	SaMST e15	LBM-Base
ER→Min	0.607	0.566 / 0.387	0.863 / 0.644	0.710 / 0.498
ER→Ukiyo	0.663	0.571 / 0.321	0.748 / 0.195	0.724 / 0.396
Imp→Min	0.602	0.623 / 0.585	0.892 / 0.731	0.823 / 0.483
Roc→Ukiyo	0.515	0.539 / 0.542	0.582 / 0.604	0.599 / 0.295

split. The same bootstrap packet also keeps SaMST strictly above IDT, but only in the same high-LPIPS region summarized in Table 1. We use this bootstrap only as within-split stability evidence; it is not presented as a cross-split or human-preference validation.

Adaptation cost and accessibility. Figure 1 makes the practical gap visible under matched same-cost budgets rather than through a synthetic normalized speedup claim. After roughly two minutes of training, the LBM step-350 row reaches transfer CLIP-S 0.6629 at LPIPS 0.3377 and sits +0.0230 above the Distinct5 IDT floor. That row is intentionally a matched-budget packet rather than the strongest retained LBM operating point: the reviewed LBM transfer frontier already appears earlier, around 1.2 minutes, with F/H/K step-1 anchors. We keep the later step-350 row on page 1 because it is the exact packet aligned to the rerun two-minute SaMST and SaMAM baselines. Retained local `nvidia-smi` traces keep those matched-budget packets within the same 8 GB RTX 4070 laptop envelope, peaking at about 5.0 GB for LBM, 6.1 GB for SaMST, and 7.5 GB for SaMAM. Under that shared budget, SaMST reaches only +0.0166 while paying a much larger LPIPS cost of 0.7487 and a slower 461 ms/image executor, and SaMAM

Table 3: Distinct5-512 same-cost records under the reviewed full 5 × 5 / 750-output packet. Train time is cumulative wall-clock to the shown checkpoint, with evaluation excluded; inference is pure 750-image generation wall time. This is recorded operating-point cost, not a normalized time-to-quality comparison.

Method	Recorded point	Train to point	Infer-750	ms/img
LBM	step 350	**1.9 min**	**84.4 s**	**113**
SaMAM	step 16	2.2 min	109.2 s	146
SaMST	40-step/style	2.0 min	345.7 s	461

Note. These are the closed same-cost rows used on page 1: all three methods were trained to roughly the same two-minute budget before evaluation. This table therefore reports the matched-budget packet, not the strongest retained LBM frontier point. The stronger reviewed LBM frontier rows still appear earlier at about 1.2 minutes (for example F/H/K step-1 anchors) and are shown separately in Figure 4. Retained local `nvidia-smi` traces keep the same-cost training packets inside the same 8 GB laptop envelope, peaking at about 5.0 GB for LBM, 6.1 GB for SaMST, and 7.5 GB for SaMAM. The SaMST row follows the original fast setting with per-style step-40 checkpoints; the table reports its combined Distinct5 packet cost rather than a normalized time-to-quality claim.

remains below IDT at -0.1383. The safe reading is therefore an evidence-closed operating-point claim rather than a universal time-to-parity theorem: within this domain-style setting, compact latent transport can shift customized stylization from a resource-intensive adaptation problem toward a minute-scale small-model learning problem. This matters beyond leaderboard optics because it lowers the barrier for small labs, independent researchers, and creators who want style-specific models without adapting a large generative prior.

Standard Benchmark: Historical strict-750

We then place the historical strict-750 packet as contextual support for the standard benchmark surface. Table 4 places LBM in a favorable compact content-preserving operating region within this reproduced standard-benchmark comparison. StyleID reaches the highest raw CLIP-style but collapses content. SaMST is marginally higher on CLIP-S and CLIP-C, yet LBM improves LPIPS and EC while also winning the artifact-sensitive metrics reported below. Relative to S2WAT and AdaIN, LBM preserves substantially more content at comparable or stronger style. Within this reproduced mixed-family standard benchmark packet, LBM is a competitive compact retrainable operating point with especially low LPIPS and strong artifact behavior; the claim is bounded to this table rather than intended as a same-interface ranking across all style-transfer regimes.

Cost matters here as deployment evidence, not as a normalized time-to-parity contest. The only evidence-closed historical row we retain is the preserved LBM record: 5.2 minutes of training to the reported checkpoint and 114 ms/image over the strict-750 protocol. Other historical baselines are omitted from the timing table be-

Table 4: Historical standard benchmark (strict-750 packet) in grouped family format. Higher is better for CLIP-S, CLIP-C, and EC; lower is better for LPIPS. Artifact-sensitive metrics are reported separately in Table 6.

Metric	CNN-based			Transformer-based			Training-free		Latent transport
	AesPA	AesFA	AdaIN	StyTr2	S2WAT	SaMST	CAST	StyleID	LBM
CLIP-S ↑	0.5976	0.6496	0.7130	0.6374	0.7139	0.7194	0.6652	**0.7597**	0.7161
CLIP-C ↑	0.5009	0.5497	0.6990	0.6001	0.7465	**0.8193**	0.5562	0.5519	0.8086
LPIPS ↓	0.8215	0.7392	0.6298	0.6345	0.5263	0.4664	0.7263	0.7497	**0.4514**
EC ↑	0.1067	0.1694	0.2639	0.2330	0.3382	0.3839	0.1821	0.1902	**0.3928**

Table 5: Historical strict-750 / 256×256 recorded operating-point cost. Training time is the wall-clock cost to the manuscript operating point; inference is the end-to-end runtime for the 750-image protocol. This is recorded operating-point context, not a normalized time-to-quality comparison.

Method	Train to point	Infer-750	ms/img
LBM	**5.2 min**	85.4 s	114

Note. The LBM row is a preserved timing record. Other historical rows are intentionally omitted here because the available train/infer evidence is either probe-estimated, mixed-protocol, or incomplete rather than a closed operating-point packet.

cause the currently available records are probe-estimated, mixed-protocol, or otherwise incomplete rather than closed operating-point packets. The practical reading is therefore narrower but cleaner: LBM remains in a compact single-image interactive regime at 114 ms/image while keeping a 3.91M-parameter footprint, and we prefer that bounded statement to a broader cross-method timing slogan built on weaker evidence.

Artifact-sensitive diagnosis

The artifact claim is clearest on the historical 256-resolution packet, where SaMST is the closest efficient baseline to LBM on the core metrics and also the closest recovered baseline with locally readable outputs for crop-based diagnosis. Table 6 combines whole-image rows with centered matched texture crops from the same reviewed packet, so the contrast is visible before and after magnification: SaMST often preserves coarse composition but pays for its stronger raw texture with muddy grain, unstable color spill, and broken local edges. LBM is less aggressive in raw style, but cleaner on the same spatial evidence.

Table 6 shows that LBM is not merely winning by leaving the image unchanged. Relative to SaMST, it improves MUSIQ from 36.10 to 49.21, MANIQA from 0.314 to 0.406, DISTS-content from 0.294 to 0.248, HF-Patch-KID from 6.76 to 4.17, and FFT slope error from 1.054 to 0.547. A paired bootstrap over the 750 per-image scores for Ours gives 95% intervals of $[0.4467, 0.4563]$ for LPIPS and $[0.7091, 0.7234]$ for CLIP-S. The photo-to-art subset ($n = 120$) yields CLIP-S 0.6716 and LPIPS 0.4882; the non-identity subset ($n = 600$) yields CLIP-S 0.6911 and LPIPS 0.4609.

Tokenizer and representation ablations

These ablations make the tokenizer conclusion sharper than a parameter-count story. Class-local prototypes and global atoms did not beat the baseline frontier, so merely enlarging the tokenizer is insufficient. Content-guided spatial routing helped, which indicates that execution location matters. The largest improvement came from a prototype-aware latent target queue, which reduces target-distribution noise before the terminal loss is applied. Finally, content-adaptive atom routing produced the best style score but regressed LPIPS, suggesting that stronger style-side separation can still demand larger executed motion. A plausible next tokenizer direction is therefore to test a two-factor interface rather than a larger undifferentiated code: a target-style carrier that remains distinguishable after execution, plus a content-risk gate that bounds how much of that carrier should be applied.

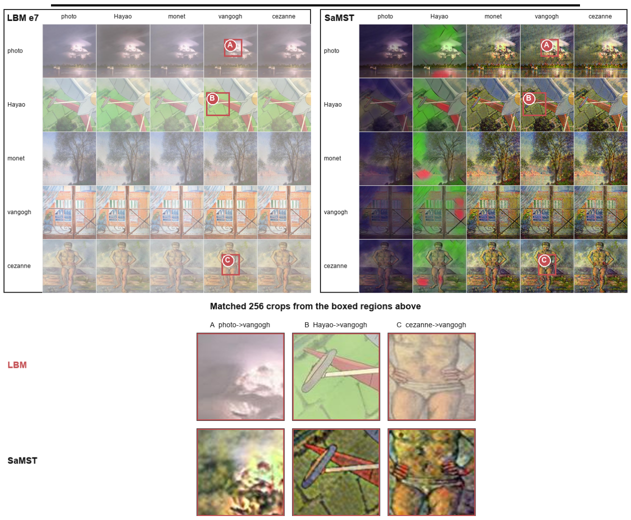
Non-training probes support the same diagnosis boundary. On the historical e8 checkpoint, tokenizer codes have near-full rank for five styles (effective rank 3.986, mean off-diagonal cosine 0.015), while generated latent residuals are much more aligned (effective rank 3.324, mean off-diagonal cosine 0.725). Residual variance is also content dominated: source-image variance explains 0.9501 of raw generated-delta variance, while target style explains only 0.0283 before source-image removal but 0.5664 after removing per-source-image residual means. On the matched Distinct5 step-1 localization packet, executor-side refresh yields the larger reviewed improvement than style-side refresh alone, which shifts immediate bottleneck suspicion toward execution rather than raw tokenizer-code weakness on that local surface. A separate matched successor probe reaches the same boundary from the execution side: tokenizer-code geometry maps only moderately into executed geometry, while realized style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. Taken together, these probes show that distinguishable tokenizer codes do not automatically remain distinguishable after content-conditioned execution. They justify evaluating tokenizer quality through executed control rather than code separability alone, but they remain local to the current localization and successor surfaces and do not prove that tokenizer-side weakness is absent.

Mechanism ablations

Removing terminal distribution matching gives excellent content preservation but weak style, showing that terminal distribution alignment is a primary style driver in the

Table 6: Artifact-sensitive diagnosis on the historical standard benchmark packet. Top: artifact-sensitive metrics among the strongest recovered methods. Bottom: a whole-image/crop composite from the reviewed LBM packet and SaMST, showing that SaMST's stronger raw texture is accompanied by muddy grain, color spill, and broken local edges, whereas LBM remains cleaner on the same source-target rows.

Metric	LBM	SaMST	S2WAT
MUSIQ ↑	49.2059	36.0950	36.5256
MANIQA ↑	0.4057	0.3139	0.1754
DISTS-content ↓	0.2477	0.2943	0.2942
HF-Patch-KID ↓	4.1694	6.7598	12.6623
FFT slope error ↓	0.5473	1.0536	0.7017
Gram micro ↓	0.0798	0.0947	0.1085



current mainline. Removing kinetic regularization preserves style but collapses content, confirming that path-energy regularization is essential. Strong color loss damages the trade-off, so naive color matching is not used in the mainline. Trajectory straightness analysis gives the same reading: the full model has PLR 0.98, while removing SA-SWD produces near-linear transport to its selected endpoint target (PLR 0.9995) and removing kinetic regularization reduces straightness to PLR 0.94 with content collapse. A landed same-family Distinct5 H packet reaches the corresponding bounded path-stability read under matched `epoch_0001` checkpoints: weakening kinetic regularization increases endpoint/path/peak motion from 80.24 / 80.28 / 80.41 in `H_base` to 111.71 / 111.72 / 111.83 in `H_k025` and 122.42 / 122.40 / 122.51 in `H_k000`, while CLIP-S / LPIPS shift from 0.6891 / 0.4272 to 0.6825 / 0.4600 and 0.6790 / 0.4862. We use this only as bounded same-family support that kinetic regularization acts as a practical path stabilizer in the current OMF / field regime, not as a broader theorem or cross-family generalization.

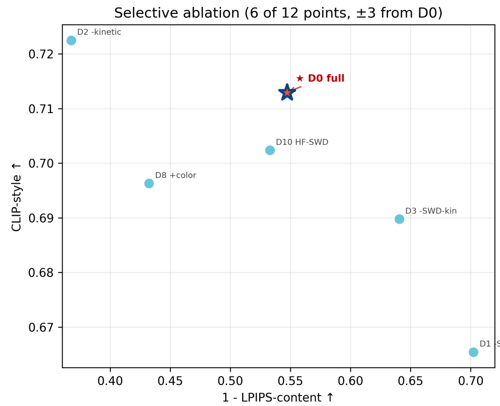


Figure 3: Destructive ablations. Removing SA-SWD improves content but weakens style, while removing kinetic regularization increases style at the cost of severe content degradation.

Sensitivity and low-cost adaptation

The 40-run category-weight sweep trained 20 sampling recipes with $K \in \{1, 2\}$ for eight epochs each, evaluating every epoch for 320 total checkpoints. The best EC result was K2 balanced default at epoch 3 with CLIP-S 0.6980, CLIP-C 0.8727, LPIPS 0.3777, and EC 0.4343. The best raw-style result in the sweep was K1 balanced default at epoch 8 with CLIP-S 0.7161, CLIP-C 0.7984, LPIPS 0.4605, and EC 0.3863. The practical reading is simple: more complex category weighting was not consistently better than balanced sampling, so we keep this sweep as a supporting sensitivity result rather than spending main-figure budget on it.

Step-count sensitivity is also stable. Evaluating 1, 4, 8, 12, and 16 integration steps on the selected epoch-7 checkpoint changes all-pairs CLIP-S only from 0.7159 to 0.7162, while LPIPS remains near 0.4514. Four steps already recover nearly identical quality, with measured strict-750 throughput increasing from 8.43 images/s at one step to 9.54 images/s at four steps and remaining around 9.34–9.50 images/s for 8–16 steps. A 3×3 grid over $\lambda_{\text{kin}} \in \{0.5, 1.0, 2.0\}$ and $\lambda_{\text{swd}} \in \{10, 20, 30\}$ produced a smooth style-content surface: higher kinetic weight and lighter terminal pressure improve content, while weaker kinetic weight or stronger terminal pressure increases style at the cost of LPIPS.

All reported evaluations use fixed seeds, stored resolved configurations, source snapshots, checkpoints, and per-epoch logs. Historical timings were measured on an RTX 4070 Laptop GPU unless otherwise noted. For Distinct5-512, Table 3 records the evidence-closed operating-point times and Figure 4 preserves their same-scope timing provenance with faster points placed to the right and evaluation excluded from the recorded train wall. For the historical packet, training and inference cost are reported explicitly in Table 5 rather than being folded into a single efficiency slogan. Across both packets, the intended takeaway is practical rather than leaderboard-style: meaningful style-id customization can remain in a minute-to-hour consumer-

Table 7: Distinct5-512 tokenizer and representation ablations. The results show that capacity alone is not the issue; target selection, content-guided execution, and sparse routing determine whether the representation is executable.

Variant	Best epoch	CLIP-S	LPIPS	Interpretation
Direct atom residual baseline	8/1	0.6876	0.4468	weak baseline; simple residual atoms do not solve representation
Class-local prototypes	8/1	0.6849	0.4464	more per-class capacity, no better style geometry
Global VQ atoms	8	0.6873	0.4446	small LPIPS gain, no style breakthrough
Content-guided spatial routing	2	0.6907	0.4226	first simultaneous style and LPIPS gain
VQ + content-guided routing	1	0.6898	0.4156	stronger content preservation, style still limited
Prototype-aware latent target queue	1/3	0.6973	0.3331	major jump from better internal target distribution
Prototype queue (base schedule)	1	0.6969	**0.3186**	current low-displacement reference point
Prototype queue (balanced schedule)	2/1	0.6994	0.3213	current balanced point
Prototype queue + adaptive routing	1	**0.7010**	0.3623	style boost, but larger endpoint movement

Table 8: Destructive ablations under the historical standard benchmark protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o SA-SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

GPU regime without adapting a large pretrained generator. An evidence-grade timing audit with explicit `closed` / `partial` / `estimated` status labels is included in the supplementary timing appendix.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. On the historical standard benchmark, StyleID has the highest raw style but collapses content, SaMST is slightly stronger than LBM on CLIP-S and CLIP-C but shows grain-like local artifacts, and LBM occupies a cleaner LPIPS/artifact-sensitive operating point. Distinct5-512 sharpens the same lesson further because its no-op baseline is measured explicitly: higher raw style is achievable, but only by moving into a severe LPIPS and targetwise ArtFID failure region, and some baselines fail even to exceed the identical-image prior. This exposes a community-relevant blind spot in art-to-art evaluation: preserved art-domain priors can masquerade as target-style success unless the unchanged-image floor is reported. Earlier AST evaluation work already notes that protocols are inconsistent across papers and that automatic scores can be style-dependent or misaligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our claim is therefore concrete but local: on Distinct5-512, the unchanged-image reference and Δ_{idt} act as useful diagnostics because they agree with the representative-direction sanity check in Table 2 and with the observed failure of sub-IDT baselines to move consistently toward the requested target style. The practical implication is separate from that metric point: the same experiments show that meaningful style-id customization can be trained and deployed in a lightweight small-model regime, lowering the barrier for researchers and creators who want customized stylization without maintaining a large generative backbone. We therefore treat adaptation cost as community-relevant evidence rather than an appendix-side detail and encourage future domain-style papers to report it alongside quality metrics.

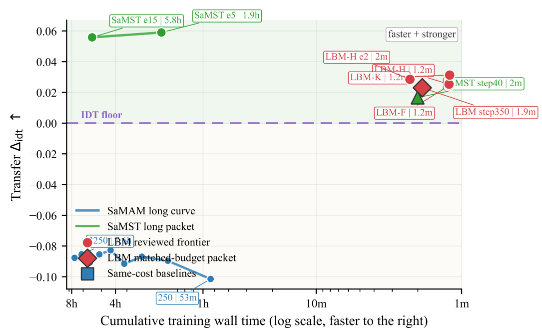


Figure 4: Evidence-closed Distinct5 timing records plotted as transfer-only Δ_{idt} against cumulative training wall time under the reproduced full 5×5 / 750-output protocol. The horizontal axis is log-scaled and inverted so faster adaptation appears to the right. LBM is shown in two roles: the retained reviewed frontier points around 1.2 minutes and the later step-350 matched-budget packet used on page 1. SaMAM is shown both as the exact same-cost step-16 packet and as bounded pre-2250 timing records with the active manuscript boundary fixed at the trusted 2250 checkpoint; SaMST is shown both as the exact same-cost step-40/style packet and as the longer retained operating point. The plot therefore preserves the calibration behind the page-1 summary rather than serving as a closed normalized time-to-parity claim.

We also bound the scope of the quality claim deliberately. Large diffusion-prior stylizers such as StyleID, ArtBank, SMS, and StyleSSP (Chung, Hyun, and Heo 2024; Zhang et al. 2024; Jiang et al. 2025; Xu et al. 2025) operate with different conditioning signals, backbone sizes, and inference budgets, and we do not claim a universal win over that regime. Our claim is narrower and, for practical customization, more relevant: among compact, efficient domain-level style-id models on the reproduced packets studied here, LBM offers a stronger measured quality-cost trade-off and a substantially lower adaptation barrier.

The tokenizer experiments narrow the design question. Within the tested Distinct5 design space, raw code capacity alone is not the bottleneck. The sharper issue is whether target-style information survives execution through a content-conditioned vector field strongly enough to yield real style gain beyond the unchanged-image reference. The current evidence therefore favors target-style carriers, prototype-aware target queues, and content-aware routing over undifferentiated token expansion. Matched successor probes reinforce the same point: tokenizer code separation maps only moderately into executed separation, while style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. This keeps executed style survival as the sharper mechanism question in the current successor family, and it suggests that the next representation stage should factor target representation from execution budget rather than simply adding parameters.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher resolutions may need better solvers or stronger constraints. Third, SA-SWD is an adaptive-projection distribution loss; current controls show that most of its benefit comes from the distributional term rather than from the specific semantic projection rule. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. Finally, Distinct5-512 is a five-domain stress split rather than a universal art benchmark, so broader domain coverage remains future work even though the current cross-method comparison is already informative on this split; we therefore do not claim universal adoption of IDT-style diagnostics without broader validation.

Conclusion

We presented Latent Bridge Matching, a compact latent transport design for domain-level artistic transfer. The method separates training-side endpoint selection, pathwise field learning, adaptive-projection terminal distribution matching, and style-tokenized execution. Under the reproduced historical standard benchmark, LBM delivers a favorable balance of style strength, content fidelity, artifact control, and practical footprint within the mixed-family comparison packet considered here. The Distinct5-512 stress benchmark sharpens the paper's main evaluation point: once the identical-image prior is made explicit, LBM stays above the no-op floor and occupies a favorable positive-Δ_{idt} CLIP-S / LPIPS region among reproduced compact operating points, while lower ArtFID alone does not guarantee useful target-style movement. On this split, reporting the IDT floor together with Δ_{idt} is a useful diagnostic rather than an optional extra statistic. Just as importantly, the reproduced minute-scale consumer-GPU adaptation path and the 3.91M / 114 ms historical operating point show that customized stylization can remain a small-model learning problem rather than a large-prior adaptation problem, lowering the barrier for researchers and creators who want low-cost style-specific models on consumer GPUs. The tokenizer study then isolates the remaining mechanism question. Within the tested Distinct5 tokenizer variants, matched localization controls show that executor-side refresh recovers more than style-side refresh alone on the current local surface, while the matched same-family path-stability packet supports the bounded claim that kinetic regularization acts as a practical stabilizer in the current OMF / field regime. The next representation target is therefore to preserve target-style separation through execution itself, via explicit target carriers and tighter content-risk control.

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Supplementary Timing Audit

Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides bounded formal analysis with proofs and empirical checks: Theorem 1 and Theorem 2 analyze the active field/solver regime through a path-action decomposition for kinetic regularization and an $O(1/K)$ Euler discretization bound. A supplementary historical note covers an earlier OT-style selector only for family-level background and is not used to justify the current headline claims. These results justify the active design without claiming equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **No**.
- All source code required for conducting and analyzing the experiments is included in a code appendix: **No**.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial**.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Partial**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.

Table 9: Distinct5-512 timing audit from retained frontier and same-cost artifacts. Train and eval times are cumulative wall-clock minutes to the shown packet, while inference is shown only when a packet-bound pure-generation log is retained. This table is an evidence audit, not a shared normalized efficiency ranking.

Method	Point	Train min	Eval min	Infer ms/img	Tr ArtFID	Status	Evidence note
LBM-LPIPS frontier	F step 1	**1.2**	0.59	–	126.8	closed	Formal remote b44 packet; strongest retained transfer $1 - \mathrm{LPIPS}$ anchor.
LBM-Style frontier	K step 1	**1.2**	0.60	–	162.0	closed	Formal remote b44 packet; strongest retained transfer CLIP-S anchor.
LBM same-cost	step 350	1.9	0.71	112.5	–	closed	Exact page-1 matched-budget packet. This row is later than the retained 1.2-minute frontier and is used only because it matches the rerun two-minute baseline packet.
SaMAM same-cost	step 16	2.2	0.92	145.6	–	closed	Exact page-1 matched-budget packet; still below the IDT floor on transfer CLIP-S.
SaMST same-cost	step 40/style	2.0	1.00	460.9	–	closed	Exact page-1 matched-budget packet under the original fast per-style step-40 setting.
LBM-Style longer	step 4	4.5	1.69	–	–	partial	Eight checkpoints landed, but paper-facing eval is only closed through step 4 and is already worse than the step-1 style anchor on style and LPIPS.
SaMST long packet	e15	347.3	2.62	–	444.5	partial	Paper-facing saturated point, but same-run-root inference timing is missing.
SaMAM long packet	2250	458.6	0.79	–	148.2	manuscript-valid	Trusted active-manuscript anchor; still below the IDT floor on transfer CLIP-S.
SaMAM audit-only	3000	612.6	4.82	–	394.8	audit-only	Closed audit packet outside the active manuscript boundary; excluded from paper-safe positive-IDT claims.

Table 10: Historical standard benchmark / legacy256 timing evidence quality. Only the LBM row is a fully closed operating-point packet; the remaining rows are retained only to explain why they are excluded from a closed shared timing table.

Method	Train to point	Infer-750	ms/img	Grade	Reason
LBM	5.2 min	85.4 s	114	actual	Preserved standard-benchmark operating-point packet.
SaMST	67.7 s/epoch probe	–	–	estimated	Historical train time is only extrapolated from a one-epoch probe.
S2WAT	$\sim$176.7 min	–	–	estimated	Train wall is only a rough $5.3\ \mathrm{s} \times 2000$ estimate; strict infer-750 is not separately measured.
StyleID	–	603.3 s / 3016.3 s est.	804 / 4022 est.	unfair packet	The recorded 603.3 s run is not a fair full-750 packet, while 3016.3 s is only an estimate.
Distinct5-256 relabels	–	–	–	mixed-protocol	Repo timing rows labeled `Distinct5-256` collapse to `legacy256_overfit50` rather than a first-class dataset surface.